

# Yuta Oshima

PH.D. STUDENT · THE UNIVERSITY OF TOKYO

Engineering bldg. 2nd 9F 92C1, GCI Matsuo Lab. 7-3-1, Hongo, Bunkyo-ku, Tokyo, JAPAN, 113-0033

☎ (+81) 90-3523-0449 | ✉ [yuta.oshima@weblab.t.u-tokyo.ac.jp](mailto:yuta.oshima@weblab.t.u-tokyo.ac.jp) | 🌐 <https://shim0114.github.io/> | 🎓 Yuta Oshima

## Summary

I'm a first-year PhD student at The University of Tokyo, advised by Yutaka Matsuo, and closely collaborated with Hiroki Furuta.

My research is driven by a passion to understand and control the worlds modeled by deep generative models. My ultimate goal is to build real-time, interactive world models that can seamlessly manifest human imagination into reality.

To move toward this goal, my research focuses on video diffusion models and world models, with recent work on alignment and memory mechanisms for long-horizon, interactive video generation.

## Education

### The University of Tokyo

PH.D. IN COMPUTER SCIENCE AND ENGINEERING

Advisor: Prof. Yutaka Matsuo

Tokyo, Japan

Apr. 2025 - Present

### The University of Tokyo

M.S. IN COMPUTER SCIENCE AND ENGINEERING

Tokyo, Japan

Apr. 2023 - Mar. 2025

### The University of Tokyo

B.S. IN COMPUTER SCIENCE AND ENGINEERING

Tokyo, Japan

Apr. 2019 - Mar. 2023

## Selected Publications

### CONFERENCE

#### Inference-Time Text-to-Video Alignment with Diffusion Latent Beam Search

YUTA OSHIMA, MASAHIRO SUZUKI, YUTAKA MATSUO, HIROKI FURUTA

NeurIPS 2025

2025

#### ADOPT: Modified Adam Can Converge with Any $\beta_2$ with the Optimal Rate

SHOHEI TANIGUCHI, KENO HARADA, GOUKI MINEGISHI, YUTA OSHIMA, SEONG CHEOL JEONG, GO NAGAHARA, TOMOSHI IYAMA,

MASAHIRO SUZUKI, YUSUKE IWASAWA, YUTAKA MATSUO

NeurIPS 2024

2024

### CONFERENCE WORKSHOP

#### AKITalk: Audio-Implicit Keypoints for Identity-Preserving Talking-Head Video Synthesis

RIKU TAKAHASHI, RONGZHI LI, YUTA OSHIMA, SHO KUNO, RYUGO MORITA, ISSEY SUKEDA

3rd Place in The ACM Multimedia Asia 2025 Grand Challenge

MMAAsia 2025 Workshop

2025

#### SSM Meets Video Diffusion Models: Efficient Video Generation with Structured State Spaces

YUTA OSHIMA, SHOHEI TANIGUCHI, MASAHIRO SUZUKI, YUTAKA MATSUO

ICLR 2024 Workshop

2024

#### Tactile In-Hand Pose Estimation through Perceptual Inference

TATSUYA KAMIJO, TOMOSHI IYAMA, YUTA OSHIMA, GENTIANE VENTURE, TATSUYA MATSUSHIMA, YUTAKA MATSUO, YUSUKE

IWASAWA

IROS 2023 Workshop

2023

Spotlight Presentation

### PREPRINT

#### WorldPack: Compressed Memory Improves Spatial Consistency in Video World Modeling

YUTA OSHIMA, YUSUKE IWASAWA, MASAHIRO SUZUKI, YUTAKA MATSUO, HIROKI FURUTA

Preprint

Dec. 2025

Work Experience

- EQUES Inc.**  
AI RESEARCHER  
Research on anime-style video generation within the GENIAC project, supported by Japan's Ministry of Economy, Trade and Industry (METI).

Tokyo, Japan  
Jul. 2025 - Present

**World-leading Innovative Graduate Study Program Co-designing Future Society (WINGS-CFS), The University of Tokyo**  
RESEARCH ASSISTANT

Tokyo, Japan  
Dec. 2023 - Present

Academic Honors

AWARDS

- 2025

**3rd Place**, The ACM Multimedia Asia 2025 Grand Challenge: Multimodal Multiethnic Talking-Head Video Generation

Kuala Lumpur, Malaysia

2024

**Outstanding Presentation Award**, JSAI Annual Conference

Shizuoka, Japan

2023

**Spotlight Presentation**, IROS 2023 Workshop on World Models and Predictive Coding in Cognitive Robotics

Detroit, U.S.A

GRANTS

- 2025

**1,000,000 JPY**, Leadership Development Program for Ph.D. (LDPP), the University of Tokyo

Academic Activities

- 2026

**Reviewer**, The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- 2026

**Reviewer**, International Conference on Learning Representations (ICLR)